

Nishant Patel

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Academic Qualifications

Year	Qualification	Institute	CPI/%
2021 - Present	B.Tech (EE)	Indian Institute of Technology Kanpur	8.5/10
2020	Class XII (CBSE)	Blooming Dale School, Budaun, UP	97.6%
2018	Class X (CBSE)	Blooming Dale School, Budaun, UP	95.0%

Scholastic Achievements

- Received the **Academic Excellence Award** *twice* for exceptional academic performance in 2021 and 2021-22 academic session
- Scored **99.7 percentile** and secured **All India Rank 3327** in **JEE Mains 2021** amongst 1.3 million candidates
- Ranked within the **top 0.1% of 1.6 million candidates** in class X, scoring full marks in social studies

Research Project

- Decentralised drone SWARMS and advanced UAV technology** 🤖 (May'23 - July'23)

Mentor: Prof. Ketan Rajawat, Department of Electrical Engineering, IIT Kanpur

Objective	• To increase the efficiency and usefulness of SWARM drones by addressing real life physical challenges
Approach	• Designed and implemented collision avoidance and congestion control algos with 88.4% accuracy • Optimized the setup of SWARM using components like Raspberry Pi , Alfa Tube, GPS • Examined all the technical requirements and specifications for each component, developed a plan to organize wiring, mount parts, and optimize system interaction , thereby enhancing the design of SWARM drones
Outcome	• With 81.2% accuracy , successfully carried out Leader-Follower tests for SWARM drones in group of 4

Key Projects

- Financial Analysis of NIFTY 50 Companies** (May'23 - ongoing)

Mentor: Prof. Suman Saurabh, Department of Industrial and Management Engineering, IIT Kanpur

Objective	• Data driven Analysis to compare and choose companies on the basis of financial ratios
Approach	• Developed an interactive NIFTY 50 firms R-Shiny dashboard , utilizing company financial statements • Scrapped 50 companies' data in R to get financial statements (balance sheet, income, cashflow statement) • Designed R code to calculate financial ratios using scraped data and generate comparative plots for analysis • Enhanced dashboard for users to visualize market response using company information and output plots.
Outcome	•With 95%+ accuracy , users can compare or judge the financial health of stock(s) for given years

- Business Analytics Using Python** | Summer Project 🐍 | Finance Analytics Club, IIT Kanpur (June'22 - Aug'22)

Objective	• To extract max value from a firm's data assets to help them give insights that yield better, faster decisions.
Approach	• Learnt concepts of data analytics and manipulation, its's representation, Statistics and Machine learning. • Created different models for price optimization and demand forecasting in Python using Price elasticity
Outcome	•Collaborating with an IOT startup Mobitra to clean and analyse data,identify trends for fleet owners

- Optimizing Travel Itineraries With Machine Learning** | Self Project 🐍 (June'23 - July'23)

Objective	• To optimize delivery vehicle travel routes using Machine Learning predictions for improved efficiency
Approach	• Utilized XGBoost model for complex numeric and categorical features, working over 120M+ data points • Experimented with LightGBM model for faster processing of categorical data with weather inputs • Implemented Genetic algorithm for taking predictions and construct time-efficient trip itinerary
Outcome	•With 93.2%+ accuracy , XGBoost model estimated a single trip's duration with error of 4.8 minutes only

- Car Parking System Using Verilog** | Self Project 🐍 (Jun'22 - Aug'22)

Objective	• To develop a robust entrance system for a parking facility using sensors to detect incoming vehicles accurately
Approach	• Incorporated password authentication for gate access, bolstering protection against unauthorized entry • Enabled prompt password input upon vehicle detection, ensuring seamless gate access control • Dynamic Gate Locking: System locks gate when another vehicle approaches, ensuring smooth traffic flow • Password-Protected access for second vehicle entry, enhancing security and gate congestion prevention
Impact	• Successfully implemented the verilog code, obtained the simulation waveform for the car parking system

- Matlab Projects** | Self Project (June'23 - July'23)

- ▷ Learned basics of Matlab from Matlab Onramp course by Mathworks and completed its accompanying project
- ▷ Learned basic image processing using Matlab from Image Processing Onramp and completed its projects

- Python and it's common uses** | Summer Project: ACA, IIT Kanpur 🐍 (Jun'22 - Aug'22)

- ▷ Learned basics of Command Line Interface and its uses such as **SSH(Secure Shell)** and related libraries like **Argparse**
- ▷ Studied about **Web Scraping and API's** and applied it to scrap the data of Professors and Students at IITK.
- ▷ Learned about **Django Framework** and its web application to create The backend of Login system.

- Data Science With R** | Stamatics Society 🐍 | IIT Kanpur (May'23 - July'23)

- ▷ Learnt and familiarized with the basics of R language and used it for **web scrapping** and analysing data through graphs.
- ▷ Built R shiny apps using **RStudio**, data visualization using Histograms, boxplots, scatterplots and used **imager library**.

Positions of Responsibility

- Assistant Placement Coordinator, Student’s Placement Office, IIT Kanpur** (Apr’23 - Present)
*Previously worked as **Company Coordinator*** (June’22 - March’23)

Leadership	- Coordinating and Facilitating placement process for 1400+ students and internship process 700+ students. - Spearheading a team of 80+ members handpicked responsibly for coordinating the internship drive. - Developing corporate liaisons with 50+ corporate giants for Internship and Placement drive 2023-2024.
Initiative	- Invited Startups for recruitment to increase success rate of internships,placements and enhance startup culture. - Introduced a series of Internship and Placement Insights together with mock tests for the benefit of students - Managed a unified database of 4500+ companies along with placement and internship portal of all students - Managed Samvardhan career fest, coordinating participation of 15+ companies to facilitate networking
Impact	- Recorded highest number of offers(170+) made till date, on Day 1 of campus internship drive 2022-2023 - Attuned Recruitment drive, securing 1200+ placement and 450+ internship offers as company coordinator - Succesfully attracted more than 5 new Startups to register for the Placement season 2023-2024
- Student Guide | Counselling Service** (August’22 - Present)
 - Guided and mentored **6 freshmen** students in acclimatizing to the Environment of the Institute.
 - Coordinated with the Counselling Service and helped in the organisation of **Orientation 2022** of **1200+** freshmen.
 - Helped them with their academics by arranging meetings with their mentors and still maintain a good rapport with them
 - Engaged as an academic mentor, provided guidance and assistance to **60 students** in **PHY** and supporting their learning
- Events and Content Secretary | Electrical Engineer’s Association** (August’22 - Present)
 - Organised various events related to Electrical Engineering department in campus
 - Conducted sessions for freshers of the Electrical Engineering Department and helped them understand the template and courses, regarding internships and research opportunities in college
 - Organized the Farewell of the batch of 2019-2023 with a team of 20 secretaries
- Events and Media Secretary , Games and Sports council** (Jun’22-Apr’23)

Leadership	Successfully managed large-scale sports events, targetting 2000+ students, through efficient coordination
Initiatives	- Initiated and fostered relationships with diverse sponsors, including Red Bull, for major sporting events - Managed and curated content for council handles, driving a remarkable 34.7% growth in engagement - Coordinated Inferno, a multi-sport hall level tournament, overseeing 500+ participants and 15+ sports - Facilitated diverse virtual and in-person events with 3k+ attendees, ensuring remarkable engagement
- Secretary | Photography Club, IITK** (Jul’22 - Mar’23)
 - Mentored and instructed the new batch in the field of Photography. Delivered engaging lectures at Pixel, the flagship event of photography club
 - Managed the coverage of various institute events like Antaragni, Udghosh, Freshers’ Night and Galaxy
 - Managed the alumni fundraising initiatives, securing over INR 4.5 lakhs in contributions to support the club’s events
 - Organized 10+ workshops competitions.collaborating with global photographers for professional exposure to the community

Relevant Courses

*Online course **Ongoing courses

Introduction to Electrical engineering	Probability and Statistics	Fundamentals of Computing
Data Structures and Algorithm*	Introduction To Electronics	Linear Algebra and Ordinary Diff. Equations
Signals,Systems and Networks	Control Systems Analysis	Introduction To Economics
Image Processing**	Microelectronics-I	Complex Variables and Partial Diff. Equations

Technical Skills

- ▷ **Programming:** C/C++, ROS, HTML ,Python ,LaTeX, R
- ▷ **Utilities and Softwares:** AutoCAD, Git, ArduPilot, Gazebo
- ▷ **Libraries and Frameworks:** NumPy, Pandas, Matplotlib, OpenCV

Extra Curricular

Cultural	- Secured 3rd position among 22 IITs in Photostory competition at Inter-IIT Cultural Meet’23 - Led 300+ juniors who participated in Galaxy’23 (Inter-Hall Cult Championship), as Hall 12 Captain
Volunteer Work	- Reached out to 200+ institutes across India to boost Paticipation by 85% in Techkriti’s events Unmukt IITK - Promoted diversity, inclusion on Campus, through an event, with 100+ attendees
Leadership	Secretary, Photography Club, MnC- Organised events and competitions, seeing 500+ participation Head Boy- Effectively managed and represented a student body of 3000+ and enhanced school fests
Social	- Promoted support for an Orphanage by contributing used belongings in Techkriti’s social drive Student Guide - Guided 6 freshmen students in acclimatizing to the Environment of the Institute
Competition	Ranked Second among 50+ teams in Parallel Universe competition conducted by E-Cell , IITK